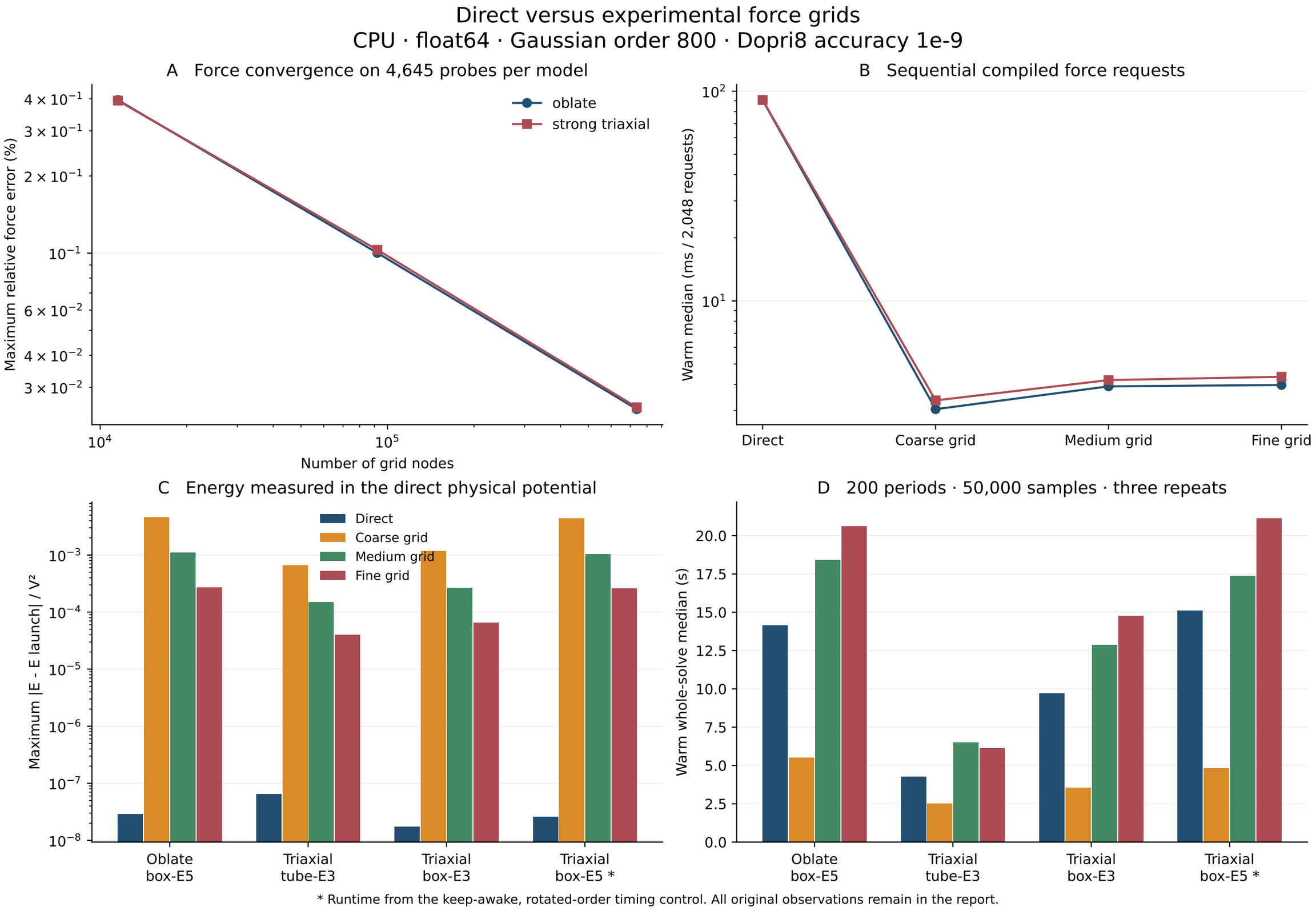
Direct versus experimental force grids
CPU · float64 · Gaussian order 800 · Dopri8 accuracy 1e-9
A Force convergence on 4,645 probes per model
Maximum relative force error (%)
oblate
strong triaxial
Number of grid nodes
B Sequential compiled force requests
Warm median (ms / 2,048 requests)
Direct
Coarse grid
Medium grid
Fine grid
C Energy measured in the direct physical potential
Maximum |E - E launch| / V²
Oblate box-E5
Triaxial tube-E3
Triaxial box-E3
Triaxial box-E5 *
D 200 periods · 50,000 samples · three repeats
Warm whole-solve median (s)
* Runtime from the keep-awake, rotated-order timing control. All original observations remain in the report.